

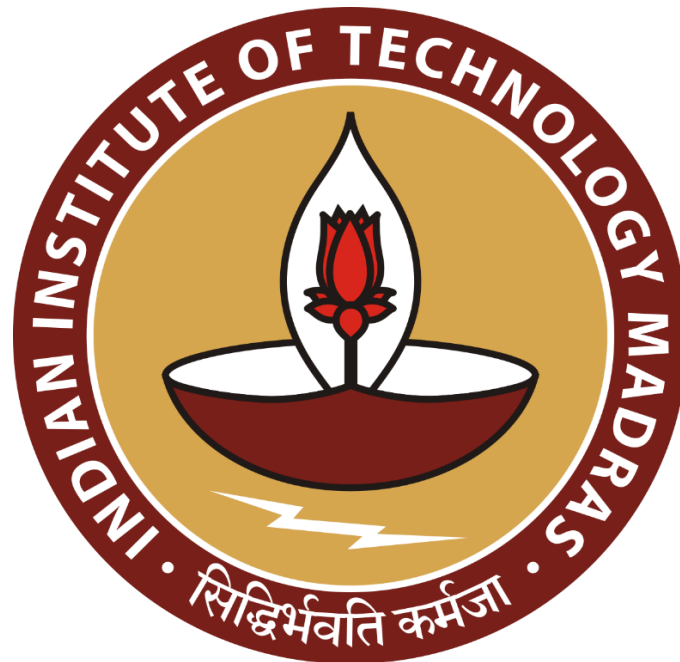
Enhancing Operational Efficiency and Driving Business Growth at JM Marketing

A Mid-Term Report for the BDM Capstone Project

Submitted by

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1. Executive Summary

JM Marketing is a leading dealer in the aluminium industry in Kolkata, offering products such as laminates, sheets, sections, and custom doors and windows. While its primary focus is on the B2B segment, the company also caters to B2C customers based on market demand.

The company faces several challenges, including delays in order fulfilment due to lengthy processing times, leading to customer dissatisfaction and attrition. This is negatively impacting sales and damaging the company's reputation in the highly competitive Chandini Market. Additionally, JM Marketing is witnessing stagnant sales growth and working capital blockage in the unsold inventory resulting in increased holding costs and thereby hampering the overall profitability of the firm.

Relevant problem-specific data was collected from the firm in Excel form, followed by a thorough data cleaning process to ensure accuracy and consistency. Initial data manipulations were then performed, setting the stage for deeper analysis. Detailed descriptive statistics were subsequently calculated to gain a comprehensive understanding of the data structure and extract meaningful insights.

The warehouse data was analysed using FSN (Fast, Slow, and Non-moving) analysis to categorize SKUs, leading to more efficient inventory management and shorter order fulfilment times. Additionally, insights were drawn from the warehouse data to identify the root cause of the firm's high working capital requirements. A comprehensive analysis of the sales data will also be conducted to uncover factors contributing to the stagnation in sales growth.

2. Proof of Originality of Data-

- Address of the firm: 37 LENIN SARANI KOLKATA-700013
- Letter from Organisation: [Click Here](#)
- Interaction with the owner Video Link: [Click Here](#)
- Collected Data: [Click Here](#)
- Analysis Data (After Cleaning and Preprocessing): [Click Here](#)



Figure 1 Inside View of business owner and one of his employees.



Figure 2 Aluminium sheets: one of SKUs



Figure 3 Racks where SKUs are stored.



Figure 4 Outside view of shop and the board of the shop

3. Metadata –

- The data is provided in Excel format (.xlsx).
- The excel workbook has two sheets: Warehouse data, Sales data
- The Warehouse data was collected from July 2022 to July 2024.
- The Sales data was collected from August 2023 to July 2024.

Warehouse data:

This data pertains to the quantity and number of pieces for each Stock Keeping Unit (SKU) stored in the warehouse, along with the date each respective SKU was received. The following details are provided:

- SKU – The name of the Stock Keeping Unit.
- Brand- The brand or company that manufactured the respective SKU.
- Family- The broader category under which similar SKUs are grouped.
- Purchased From – The name of the supplier from whom the SKU was purchased.
- Rack No: The location where the SKU is stored in the warehouse.
- Receive Date – The date on which the SKU was first received in the warehouse
- Receive Qty. - The quantity received, measured in kilograms (kgs).
- Receive Pcs. -The number of pieces of SKU received, measured in units

Sales data:

This data pertains to the quantity and number of pieces sold for each Stock Keeping Unit (SKU), along with the date on which each specific SKU was sold by the firm. The following details are provided as follows:

- SKU – The name of the Stock Keeping Unit.
- Brand- The brand or company that manufactured the respective SKU.
- Family- The broader category under which similar SKUs are grouped.
- Sale Date – The date on which the particular SKU was sold.
- Sale Qty. -The quantity sold, measured in kilograms (kgs)
- Sale Pcs. – The number of pieces of SKU sold, measured in units.

- Remark – The location from which the SKU was sold, along with any additional information required by the firm.

4. Descriptive Statistics –

Descriptive statistics for the Warehouse and Sales data was analysed using Microsoft Excel's Data Analysis Tool. Key attributes include measures of central tendency such as the Mean (average value), Median (middle value), and Mode (most frequently occurring value). Measures of dispersion include Standard Deviation (spread of data points), Sample Variance (differences from the mean), Standard Error, Kurtosis and Skewness (asymmetry). Additional statistics including Range, Maximum, Minimum, Sum, and Count were calculated thereby providing a comprehensive overview of the data statistics.

It gave a comprehensive look to the data and help in the identification of outliers present in the data.

Descriptive Statistics on Warehouse data:

In warehouse data detailed descriptive statistics was performed on attributes namely Receive Qty.(kgs), Receive Pcs., NET Qty.(kgs) and NET Pcs.

	Receive Qty.(Kgs)	Receive Pcs.	NET Qty.(kgs)	NET Pcs.
Mean	54.13077772	24.83866739	8.482903332	4.616602949
Standard Error	0.117666266	0.422361047	0.212979479	0.312531289
Median	53.72	16	0	0
Mode	55.7	12	0	0
Standard Deviation	11.25851592	40.41225009	20.37825229	29.9035451
Sample Variance	126.7541808	1633.149958	415.2731665	894.2220097
Kurtosis	1.43357949	194.867436	3.086305121	559.1939129
Skewness	-0.000366553	11.47205236	2.149728516	21.47991622
Range	91.9	840	98.79	854
Minimum	3.8	0	-3.09	-14
Maximum	95.7	840	95.7	840
Sum	495567.27	227398	77660.98	42265
Count	9155	9155	9155	9155

Figure 5 Descriptive Statistics of warehouse data

Descriptive Statistics on Sales data:

In warehouse data detailed descriptive statistics was performed on attributes namely Sale Qty. (kgs) and Sale Pcs.

	Sale Qty.(Kgs)	Sale Pcs.
Mean	46.37229274	20.50387936
Standard Error	0.223710829	0.317712794
Median	52.37	14
Mode	0	0
Standard Deviation	21.40504682	30.39267889
Sample Variance	458.1760293	923.7149303
Kurtosis	0.576866009	145.6661381
Skewness	-1.172194272	8.632784921
Range	91.1	840
Minimum	0	0
Maximum	91.1	840
Sum	424538.34	187631
Count	9155	9151

Figure 6 Descriptive Statistics of Sales data

5. Detailed explanation of Analysis Process –

Data Collection:

The structured data was collected in Excel spreadsheet from the firm containing two sheets: Warehouse data and Sales data ranging from July 2022 to July 2024 and August 2023 to July 2024 respectively.

Data Cleaning:

Corrections were made to the data due to several typing mistakes. The Receive Date was mistakenly listed as 2025 and 2026 in some entries, but after consulting with the firm, it was confirmed that the correct year should be 2024. Similarly, the Sale Date had been mistyped as

2025 in some places, which was corrected to 2024. Additionally, some SKU brand names were incorrect, and the firm requested these be corrected before proceeding with the analysis.

Net Qty (in kilograms) and Net Pcs were calculated for each SKU in the warehouse data before conducting descriptive statistics and detailed analysis. These metrics represent the total weight (in kg) and count (in pieces) of each SKU currently stored in the warehouse.

Analysis of Data:

After preprocessing the data, descriptive statistics was performed to thoroughly comprehend the dataset. These statistics offer an essential overview of key features, enabling the extraction of valuable insights. Central tendency measures, such as mean, median, and mode, illustrate the core values in the data and give a good overview of the data, while dispersion metrics, including range, variance, and standard deviation, reveal variability and distribution. Together, these metrics facilitate a deeper analysis, uncovering patterns, trends, and possible anomalies within the dataset.

A comprehensive examination of the sales data was performed, emphasizing time series analysis to track trends over various periods. This approach enabled the identification of sales patterns, seasonal fluctuations, and overall growth trends. By analysing these trends, we can recognize instances of stagnation or decline and explore their underlying causes. Such insights will assist the firm in formulating actionable strategies to tackle the issues related to stagnant sales growth. Top performing SKUs and families were identified so that firm could focus on the same to improve profitability and growth of the firm.

Brand wise sales were also analysed to understand consumer buying preference and help the firm to adapt to consumer demand and brand preference.

FSN (Fast, Slow, and Non-moving) analysis was applied to warehouse data to categorize SKUs (Stock Keeping Units) according to their movement rates. This classification is vital for effective inventory management, highlighting fast-moving items that should be prioritized for restocking, alongside slow-moving or non-moving SKUs that may be considered for discontinuation. Eliminating non-moving SKUs as scrap will not only help in decreasing holding costs for the firm and thereby improve overall profitability but also help to reduce stagnant working capital in the unsold stocks, thereby reducing the overall working capital requirement of the firm.

FSN analysis was considered because it clearly categorizes the SKUs based on movement, and it is known that the firm is facing operational inefficiency; thus, this analysis helps to strategically place the SKUs in the warehouse and improve order processing time and reduce customer dissatisfaction and their attrition and thereby help in improving the goodwill of the firm.

6. Results and Findings –

Sales Trend -

Sales data analysis revealed stagnant growth for the firm, with a notable spike in February due to seasonal sale as reported by the firm. However, the linear trend line shows a clear decline in overall sales performance following that spike, indicating a general downward trend in sales.

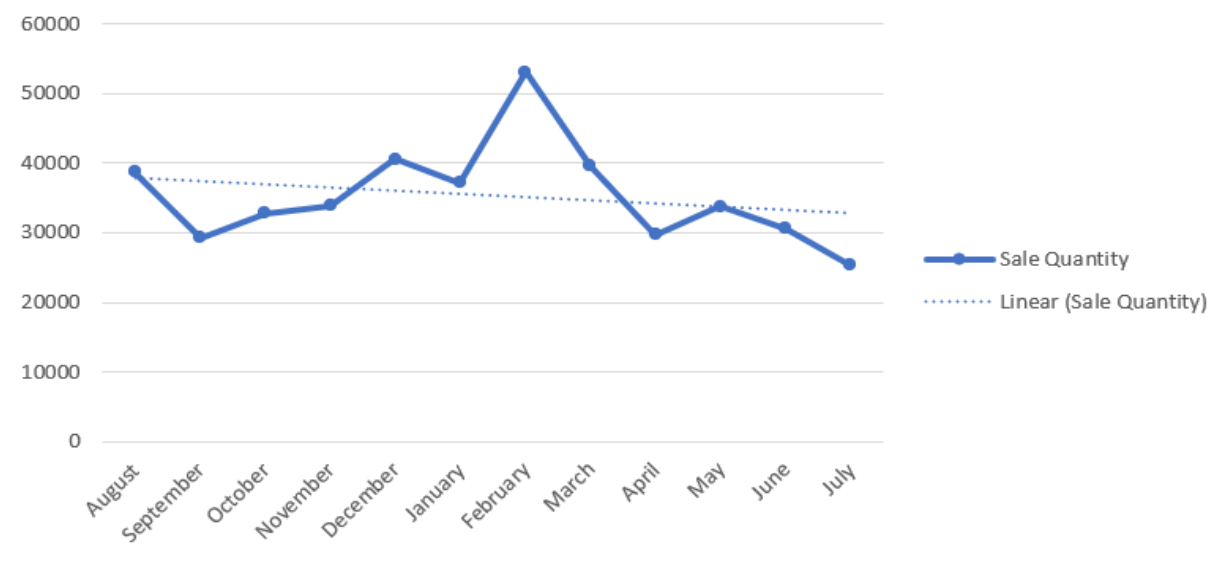


Figure 7 Sales Trend from August 2023 to July 2024.

Top five selling SKUs -

From the sales data top five selling SKUs for the firm were identified. SKUs namely RT 100*45 LOCAL 12 FT 1.5 MM, RT 50*25 JINDAL 12 FT 1.5 MM, RT 38*25 LOCAL 12 FT 1 MM, C/W 4" Screw LOCAL 16 FT 1.8 MM and RT 38*25 LOCAL 12 FT 0.5 MM were the top five selling SKUs.

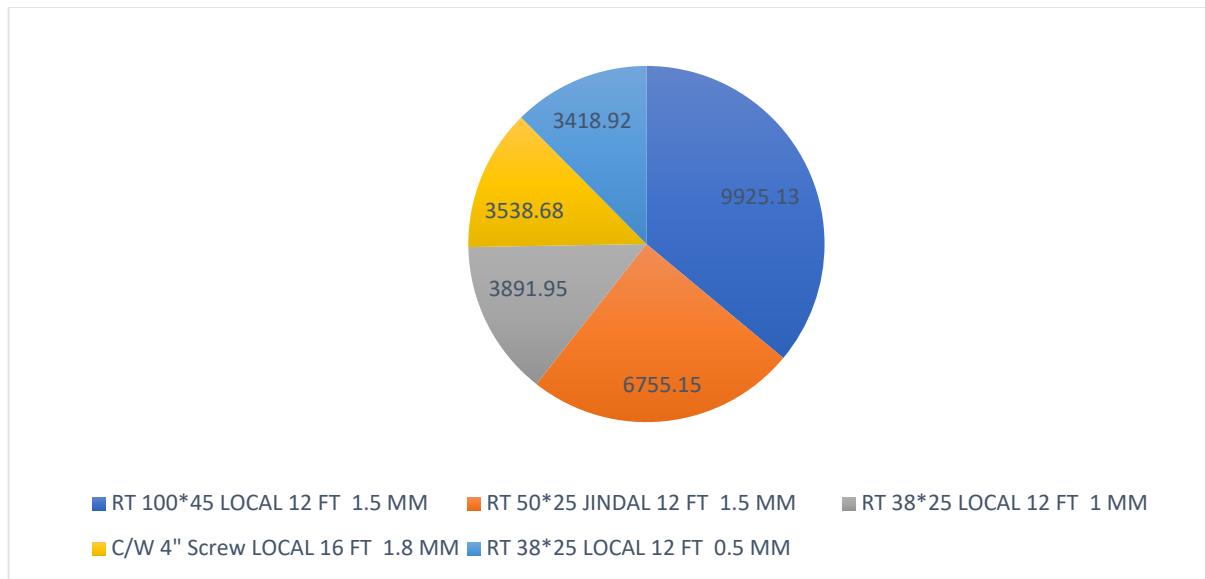


Figure 8 Top five selling SKUs

Identification of Top Suppliers –

A Pareto analysis was conducted to identify the top SKU suppliers, revealing that Vikash, Pritstino, and Kamakhya were the leading contributors.

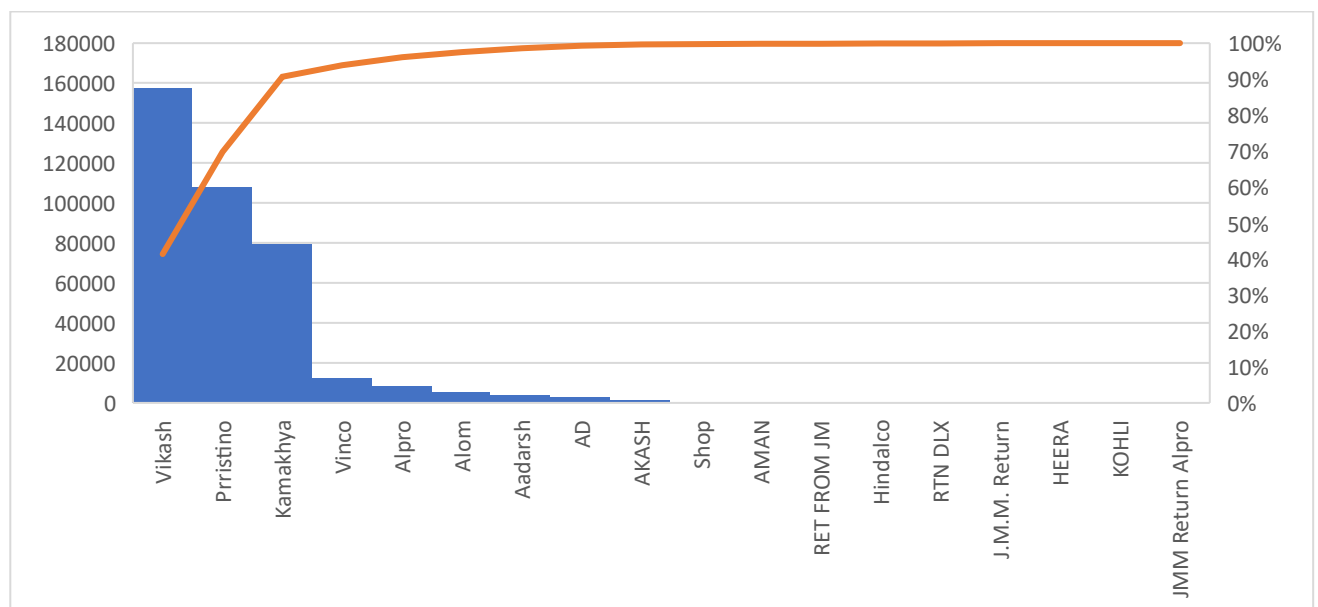


Figure 9 Pareto Chart for top suppliers of the firm